

Executive Strategy Memo · Section 1: The Situation & Top 5 Offender List

THE INFORMATION GAP: WHY CURRENT ROUTING IS FLYING BLIND

Delhivery's operational planning currently relies on **OSRM-based route estimates** — a tool that assumes frictionless roads, infinite vehicle capacity, and zero structural congestion. In practice, the network does not behave this way. Every day, **shipments stall, SLAs are missed, and cost cascades silently** through the hub network — not because of random variation, but because of *mathematically identifiable* structural bottlenecks that standard routing models are constitutionally blind to.

Our graph-topology analysis, applied across **144,867 logged trips**, has closed this information gap. By modelling Delhivery's hub network as a directed graph and computing network-theoretic risk properties — corridor chronicity, delay amplification, and structural chokepoint exposure — we have **pinpointed the exact facilities bleeding time and money from the P&L**. The five facilities below are not anecdotal trouble spots: they are the *mathematically confirmed* critical nodes whose dysfunction propagates the farthest, the fastest, across the entire network.

★ TOP 5 NETWORK OFFENDERS: FACILITY RISK REGISTER

1

GURGAON – BILASPUR HUB (Haryana)

Facility ID: IND000000ACB

Structural Chokepoint Chronic Problem Corridors Peak Delay Multiplier Score

100.0 / 100 **85 corridors** **5.45×** **over ETA**

The undisputed #1 network chokepoint. Gurgaon–Bilaspur carries the **highest network centrality of any hub in the system** — meaning disruption here ripples farthest. With 85 chronically delayed corridors and a peak delay exceeding **5×** the **OSRM estimate**, it is the single highest-priority intervention target.

2

KOLKATA – DANKUNI HUB (West Bengal)

Facility ID: IND712311AAA

Structural Chokepoint Chronic Problem Corridors Peak Delay Multiplier Score

47.7 / 100 **46 corridors** **9.94×** **over ETA**

The most **severe delay amplifier** in the Top 5. Kolkata–Dankuni posts the worst peak delay ratio in the cohort: shipments regularly take **nearly 10×** longer than OSRM predicts. While its chronic corridor count is lower, the magnitude of disruption when it fails is extreme — a systemic SLA liability for East India flows.

3

BANGALORE – NELAMANGALA HUB (Karnataka)

Facility ID: IND562132AAA

Structural Chokepoint Chronic Problem Corridors Peak Delay Multiplier Score

39.6 / 100 **68 corridors** **4.0×** **over ETA**

Bangalore–Nelamangala carries the **second-highest chronic corridor burden** in the network (68 corridors). As the primary gateway to South India's largest e-commerce market, persistent underperformance here has an outsized commercial consequence, affecting both B2C SLAs and high-value fulfillment commitments.

4

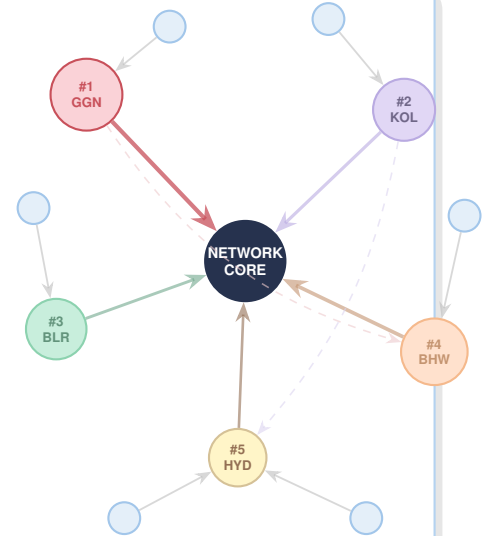
BHIWANDI – MANKOLI HUB (Maharashtra)

Facility ID: IND421302AAG

Structural Chokepoint Chronic Problem Corridors Peak Delay Multiplier Score

100.0 / 100† **58 corridors** **5.57×** **over ETA**

Bhiwandi–Mankoli records the **highest absolute SLA Impact Score** of all facilities (†raw score: 1,911 — nearly 41% above the #1-ranked hub on raw volume of delay). Serving as the primary gateway for Mumbai's

Network Bottleneck Map

Node size → network influence. Arrow weight → SLA impact score. Dashed arcs → confirmed delay propagation pathways. Red = highest-priority intervention.

Risk Scorecard at a Glance

Hub Score	Chokepoint	Chronic	Peak
Lanes			
Delay			
Gurgaon	100.0	85	5.45×
Kolkata	47.7	46	9.94×
Bangalore	39.6	68	4.0×
Bhiwandi	100.0†	58	5.57×
Hyderabad	32.6	56	9.94×

†Bhiwandi's composite risk is ranked #4 due to lower network centrality vs. Gurgaon, despite a higher raw SLA Impact Score.

